

AGENTIC CODING

Understanding autonomous AI agents in software development

WHAT IS AGENTIC CODING?

AGENTIC CODING

- AI agents that autonomously write, test, and debug code
- Agents plan multi-step tasks and iterate on solutions
- Human provides goals; agent handles implementation details



Agentic coding shifts AI from assistant to autonomous developer.

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HOW IT DIFFERS FROM COPILOTS

- Copilots suggest code; agents execute entire workflows
- Agents maintain context across files and sessions
- Agents can run tests, fix errors, and refactor autonomously



Agents own the development loop; copilots only suggest snippets.

CORE CAPABILITIES

AGENTIC CODING

- Task planning and decomposition into subtasks
- Code generation, editing, and refactoring across repositories
- Automated testing, debugging, and verification loops



Agents combine planning, coding, and validation in one loop.

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COMMON USE CASES

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- Building features from high-level specifications
- Automated bug fixing and regression resolution
- Legacy code modernization and documentation generation



Agents excel at end-to-end tasks with clear success criteria.

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CHALLENGES AND RISKS

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- Hallucinated APIs or incorrect logic in complex tasks
- Security vulnerabilities from unreviewed generated code
- Difficulty specifying precise requirements for ambiguous tasks



Human oversight remains essential for correctness and security.

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KEY TAKEAWAYS

AGENTIC CODING

- Agentic coding automates the full development cycle
- Best suited for well-defined, testable tasks
- Requires guardrails: review, testing, and sandboxing
- Rapidly evolving — tooling and best practices improve monthly



Agentic coding is a powerful multiplier, not a replacement for developers.

THANK YOU

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